located towards the upper epidermis and the phloem towards the lower epidermis.In C_4 grasses, the bundle sheath cells are living and involve in C_4 photosynthesis. This sheath is called **Kranz sheath**.

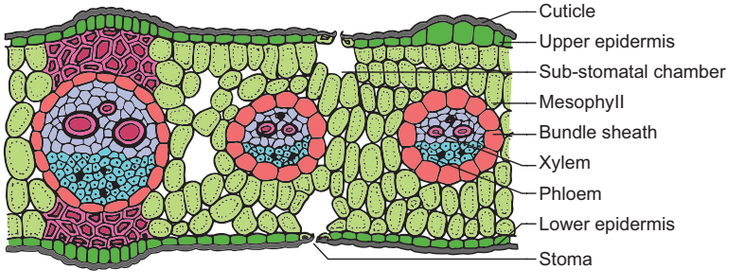


Figure 9.21: T.S. of monocot leaf (Grass)

Summary

A Tissue is a group of cells that are alike in origin, structure and function.There are two principal groups: (1) Meristematic tissues and (2) Permanent tissues. Meristematic tissues comprise of self-perpetuating cells. Meristems are classified into several types on the basis of position, origin, function and activity. Many anatomists illustrated the root and shoot apical meristems on the basis of the type and arrangement and accordingly proposed many theories. The permanent tissues normally develop from apical meristem. They are classified into two types: 1)Simple permanent tissues and 2)Complex permanent tissues. Simple tissues are composed of a single type of cells only. It is of three types: (1) Parenchyma (2) Collenchyma and (3) Sclerenchyma. A complex tissue is a tissue with several types of cells but all of them function together as a single unit. It is of two types – xylem and phloem. Secretory tissues produce different types of chemicals. Some are in the form of enzymes, hormones, rubber, gum etc.

The tissues can be classified on the basis of their function, structure and location into epidermal tissue system, ground tissue system and vascular tissue system. Epidermal tissue system develops as the outermost covering of the entire plant body. It consists of epidermal cells and associated structures. All tissues except epidermis and vascular tissues constitute the ground tissue. The vascular tissue system is formed of vascular bundles.

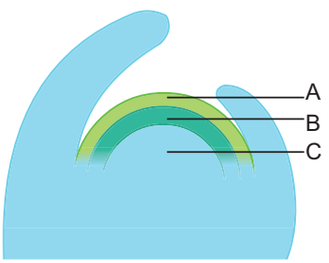
In the primary structure, the outermost layer of the root is called piliferous layer. Cortex consists of only parenchyma cells. All the tissues present inside endodermis comprise the stele. In dicot (Example: bean) root, xylem is tetrach. Its phloem patch consists of sieve tubes, companion cells and phloem parenchyma. In monocot (Example: maize) root, xylem is polyarch.

In dicot (Example: sunflower) stem, stele is eustele type and its vascular bundles are wedge shaped, conjoint, collateral, open and endarch. In monocot stem (Example: maize) vascular bundles are scattered and skull shaped, conjoint, collateral, closed and endarch.

In dicot (Example: sunflower) and monocot (Example: grass) leaves vascular bundles are conjoint, collateral and closed.

Evaluation

1. Refer to the given figure and select the correct statement.



GJ60H

   i. A, B, and C are histogen of shoot apex
   ii. A Gives rise to medullary rays.
   iii. B Gives rise to cortex
   iv. C Gives rise to epidermis

   a. i and ii only c. i and iii only
   b. ii and iii only d. iii and iv only

2. Read the following sentences and identify the correctly matched sentences.

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